

Characteristics of Trenched Coplanar Waveguide for High-Resistivity Si MMIC Applications

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Abstract—A novel low RF loss trenched coplanar waveguide (CPW) transmission-line structure fabricated using evaporated aluminum tracks on a high-resistivity ($10\text{-k}\Omega\cdot\text{cm}$) silicon (HRS) substrate is reported. By assuming that Schottky contact boundaries exist at the metal silicon substrate interface in the CPW line, the finite-element analysis method is used to determine the simulated behavior of the structure. The distributed capacitance, leakage conduction current, and dynamic shunt conductance for the line are shown to be a function of dc bias applied to the line, and also to reduce as a function of trench depth in the normal bias regime. Experimental results show: 1) the reduction of RF losses in comparison with conventional aluminum conductor CPW line structures may be as much as 0.5 dB/cm at 30 GHz; 2) by proper positive dc biasing of a CPW line on a p-type HRS substrate, a further reduction (0.2 dB/cm) in RF loss at 30 GHz can be achieved; 3) predicted trends in line leakage current, capacitance, and relative characteristics impedance are experimentally verified. The proposed waveguide structure may be utilized in a special fabrication process designed for RF/microwave applications.

Index Terms—Capacitance, conductance, CPW, loss, MMIC.

I. INTRODUCTION

MILLIMETER-WAVE frequencies are being proposed for high-volume low-cost/size/mass commercial applications. These applications include sensors for vehicle monitoring and control, radio local area networks (LAN's), secure cellular-radio systems, and short-range wide-band radio links. To achieve low cost with moderate-to-high-volume circuit production, monolithic circuitry is preferred. At microwave frequencies monolithic microwave integrated circuits (MMIC's) employ GaAs technology with integrated active devices. However, GaAs real estate is expensive, particularly when passive

circuit elements occupying large areas are required, e.g., filters, diplexers, antennas, etc.

A number of alternative substrates can be considered for planar millimeter-wave circuit or interconnect design. While GaAs offers low dielectric loss, high dielectric constant, and compatibility with MESFET and high electron-mobility transistor (HEMT) active-device technology, it is recognized that large-area GaAs substrates are costly to produce with low defect density. Silicon technology is mature, both for the production of uniformly polished high purity substrates, and for device fabrication and integration. High-resistivity silicon (HRS) offers constant thickness and high dielectric constant which is stable with frequency. In addition, polished silicon wafer substrates are very cost effective, and it has been reported that transmission lines on silicon substrate material have acceptable RF losses even in the microwave-frequency regime [1]–[4].

By employing silicon as a low-cost low-loss substrate medium, relatively large-area passive elements can be realized at millimeter-wave frequencies. Moreover, discrete silicon or GaAs devices [5] can be incorporated into these silicon circuits. This provides a flexible technology for planar circuits in which the active device most suitable for a particular task may be employed. A future extension may be the direct integration of SiGe heterojunction bipolar transistor (HBT) devices to give silicon-based circuits the capability of efficient operation at high microwave frequencies [6], [7]. The HRS substrate has been reassessed as a potential microwave material for MMIC's by Reyes *et al.* in 1995 [4] and, most recently, by Burghartz *et al.* [7]. Reyes *et al.* show that the RF dissipation loss in a coplanar waveguide (CPW) line constructed directly on an HRS substrate is one order lower than that of a CPW line with an oxide insulation layer between metal and semiconductor. In this paper, we demonstrate that a reduction in RF dissipation loss can be realized by means of trenching the HRS substrate and by dc biasing the CPW signal line. The proposed waveguide structure may be utilized in a special fabrication process designed for RF/microwave applications.

At the signal frequencies lower than 100 GHz, the transmission losses in conventional CPW lines are mainly contributed by conductor ohmic loss and dissipation loss within the dielectric substrate material. Conductor loss is influenced by

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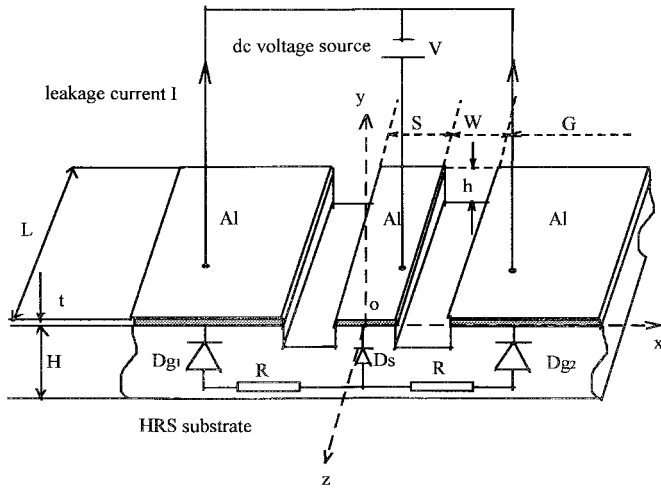


Fig. 1. Schematic diagram of the modeled trenced CPW line and its electrical equivalent circuit.

the conductivity of the metal, and the conduction current distributions on the CPW lines. Substrate dissipation loss is dependent on the leakage conduction current distribution and the electric-field distribution in the semiconductor substrate. The metal–semiconductor contacts between CPW metallization and the host silicon substrate in conjunction with different designed line configurations will lead to a variety of electric-field patterns. These, in turn, result in different conduction current distributions, different line shunt conductances, and ultimately, different RF losses.

In this paper, we illustrate the effect on electric-field, conduction current, distributed capacitance, and characteristic impedance due to vertically etched side grooves in the gaps in the CPW line between signal and ground tracks, as shown in Fig. 1. Each of these is evaluated as a function of the dc bias on the CPW signal track with respect to its ground planes. A number of theoretical predictions are presented and compared with experimental results. The effects of temperature on transmission lines based on silicon substrate have been previously reported in [7] and [8].

II. A PHYSICALLY BASED DEVICE MODEL FOR TRENCHED CPW LINES

The typical electromagnetic field distributions obtained for a conventional dielectric substrate-based CPW line by Gupta in 1981 [9] are altered by the presence of trench effects and metal–semiconductor contact effects. Consider a trenced CPW line based on HRS substrate, as shown in Fig. 1. This is constructed on (H) 650- μm -thick 10-k $\Omega\cdot\text{cm}$ p-type silicon substrate with $\langle 100 \rangle$ crystal orientation. The line dimensions are selected to give an approximate 50- Ω characteristic impedance on an untrenched line and simultaneously to accommodate 200-pitch ground–signal–ground (GSG) CPW probes. Line spacing W is 40 μm , signal track width S is 70 μm , and line length L is 6100 μm . The widths of the ground lines G are 2.5 mm. Using standard silicon-processing methods, a 1.46- μm -thick (t) aluminum metallization layer is

evaporated onto the HRS substrate (this is thinner than the thickness of 4 μm associated with three skindepth penetration on bulk aluminum for a nominal operating frequency of 30 GHz).

For most semiconductors (owing to the presence of interface states), when metal is evaporated directly onto the HRS, a metal–semiconductor Schottky barrier is formed [10]. The actual barrier height is determined by the property of the semiconductor surface and the contact metal. Schottky contacts in the simulation are modeled by the work function of the electrode metal and an optional surface recombination velocity (in this paper, values of 4.71 eV and 1.62×10^6 m/s are used, respectively). The CPW trench depths (h) examined are 0, 3, 6, 9, and 12 μm . Trenching is obtained by processing the substrate in a capacitively coupled glow-discharge plasma excited by a 70-W 13.56-MHz RF source at CF_4 pressure of 150 mtorr. The observed silicon etch rate is about 1.5 μm per hour. The silicon surface charges are assumed negligible in the trench. Based on the nominal resistivity, the p-doping density is around the order of 10^{12} cm^{-3} .

For the geometry in Fig. 1 the two-dimensional Poisson's equation, electric-charge carrier motion equations, and carrier continuity equations in the CPW line based on the HRS substrate are solved selfconsistently by the finite-element-analysis method using a commercial semiconductor device simulation package.¹ Results from the simulation are as follows.

- 1) Typical static electric-field line distributions in the trenced CPW lines are shown in Fig. 2 for different trench depths. From these, it can be seen that there are some special regions near the silicon surface where the electric line distributions are radically different from those of the classical CPW case. It is also interesting to note that the depth of this region is of the order of the silicon diode depletion region depth (10–20 μm), which has been estimated for an applied reverse bias voltage 2.0 V and a p-silicon doping concentration $1\text{--}3 \times 10^{12} \text{ cm}^{-3}$, bulk resistivity 10 k $\Omega\cdot\text{cm}$ [11]. Furthermore, as can be seen, the maximum electric fields are effectively decreased as the trenced depth increases. In each case, the dc-voltage difference between signal line and ground plate is chosen to be 2.0 V.
- 2) The corresponding leakage conduction-current contour distributions are shown in Fig. 3. As can be seen, the current density with its pattern crowding at the edges of the signal and ground lines is effectively reduced as the trencing depth increases.
- 3) Simulated distributed capacitance versus signal bias is plotted for different trenced CPW lines in Fig. 4. All of the distributed capacitances show a nonlinear response as the signal bias changes, owing to the presence of two metal–semiconductor contact layers which act as two back-to-back Schottky diodes, as shown in Fig. 1. The trend that a maximum capacitance is formed

¹ SILVACO Int. ATLAS User's Manual, Device Simulation Software, Edition 4, Oct. 30, 1996, pp. 3-2–3-54.

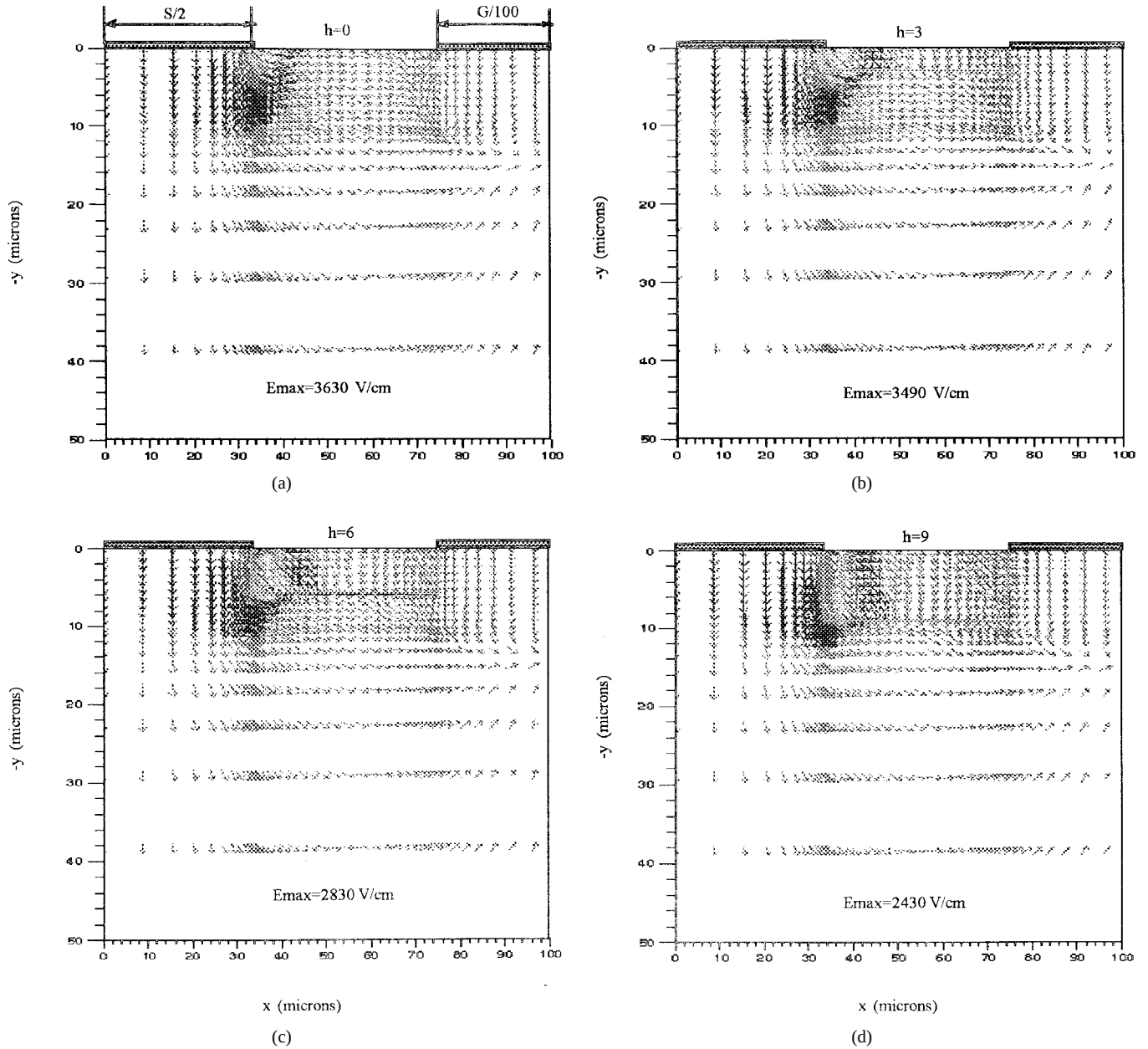


Fig. 2. E -line distributions in modeled CPW devices.

around zero-bias operation occurs, since at least one of the Schottky diodes is always in the reverse operation regime when the signal bias is moved away from the zero-bias point. Moreover, the asymmetry of CV curves with increased bias can be eliminated in the simulation by equating the ground track width G to one half of the signal linewidth.

- 4) Simulated dynamic shunt conductance versus signal bias for different trenched CPW lines is shown in Fig. 5. For negatively bias (forward) operation, the dynamic shunt conductance of the modeled CPW line is relatively larger than that of positively biased (reverse) case. This is due to the difference of two Schottky diodes. In the zero bias and reverse bias regimes, relatively small leakage currents are induced; here, the calculated shunt conduc-

tance is reduced as the trench depth increases. However, for large forward bias (GSG voltage < -3.0 V), large leakage currents are induced. Simulated results show that unlike the reduced CPW line capacitance which occurs with increased trench depth, the deeper the trench, the larger the shunt conductance.

- 5) The simulated total distributed capacitance C_t , the total leakage current I_t (electron plus hole motion current), the effective shunt conductance G_d , and the normalized characteristic impedance as function of trench depth h are listed in Table I.
- 6) Simulated leakage current versus signal bias for different trenched CPW lines is shown in Fig. 6. As can be seen, the I - V curve at different trench depth shows the forward and reverse current-voltage characteristics of a

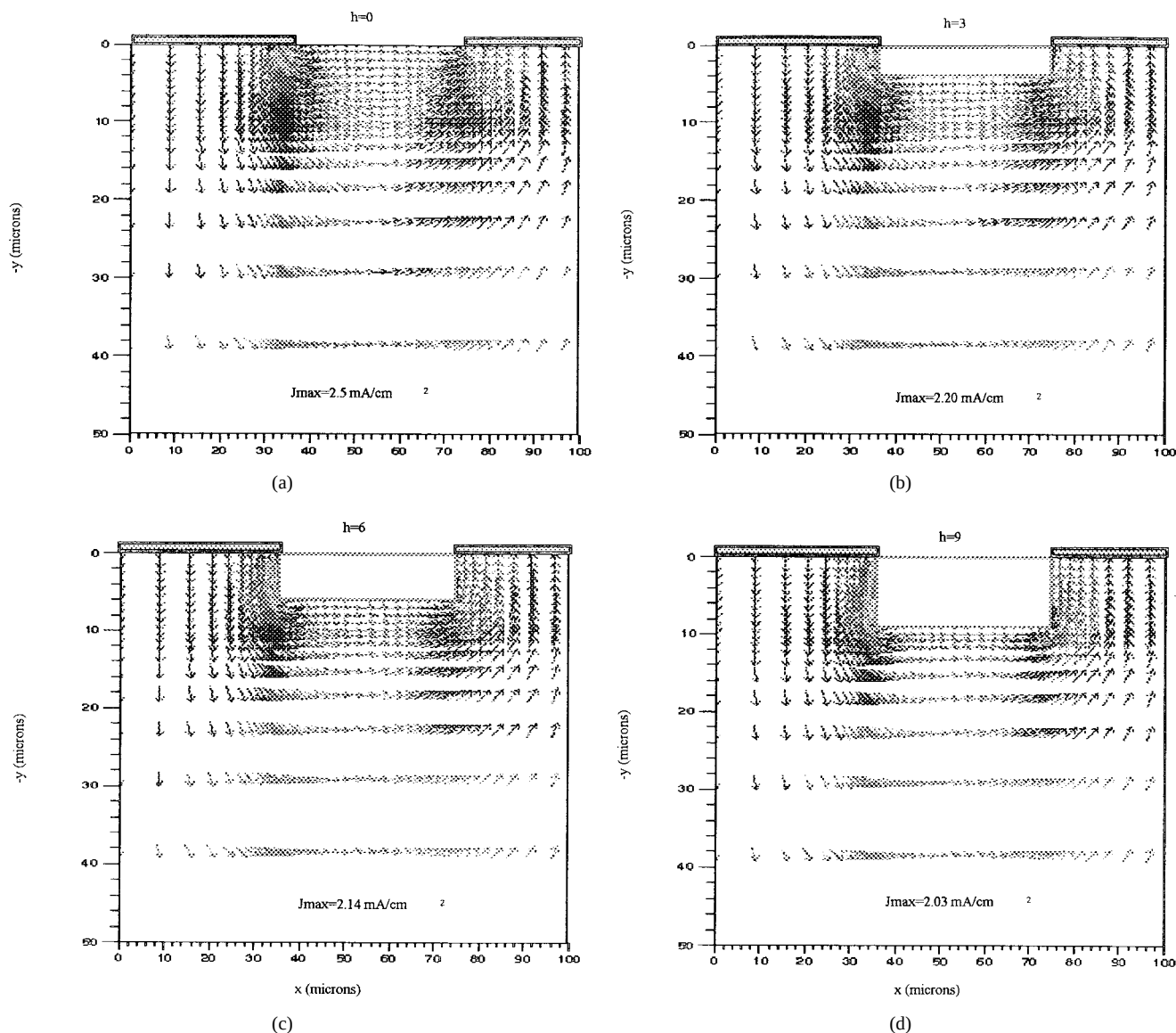


Fig. 3. Current flow patterns in modeled CPW devices.

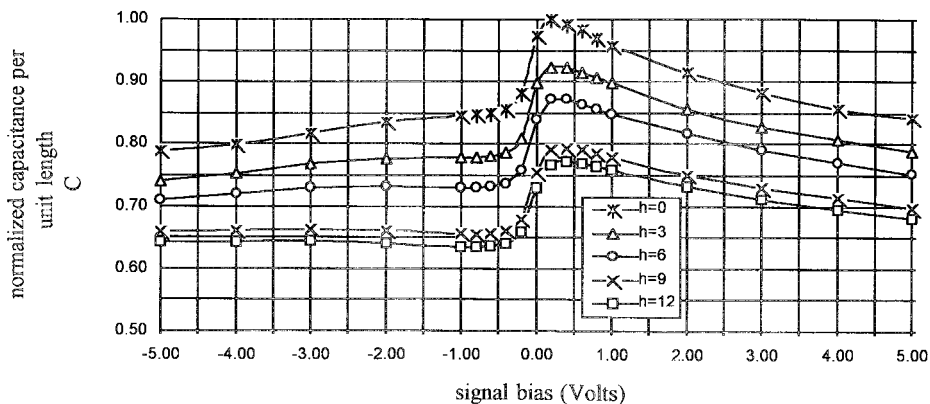


Fig. 4. Simulated distributed capacitance per unit length in trenched CPW lines as function of signal bias.

back-to-back diode pair. Both the currents for positive and negative bias are reduced due to the trenching effects.

In Table I, the electrostatic distributed capacitances are estimated using the derivative of the induced electrical charge

on the signal line with respect to variation of signal voltage, i.e., a small ac analysis approach [12]. For a nontrenched CPW line, the calculated capacitance per unit length (1.42 pF/cm) is 18% smaller than the 1.7 pF/cm (that of a conventional CPW line with an ideal dielectric constant), estimated by conformal

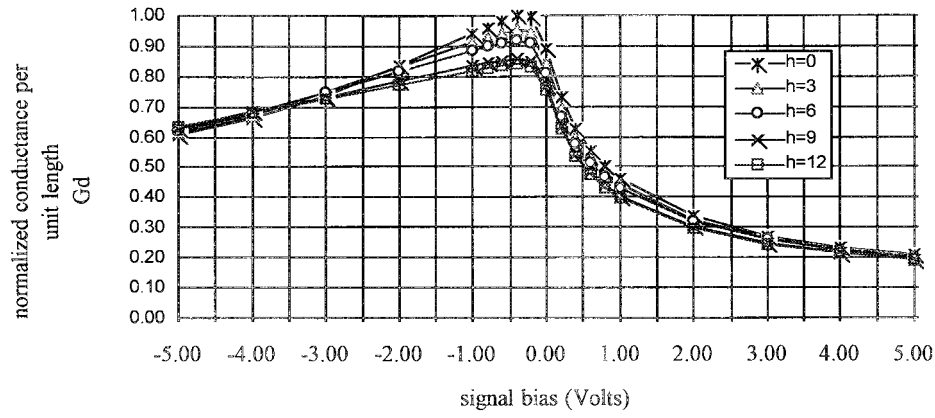


Fig. 5. Simulated shunt conductance per unit length as function of signal bias.

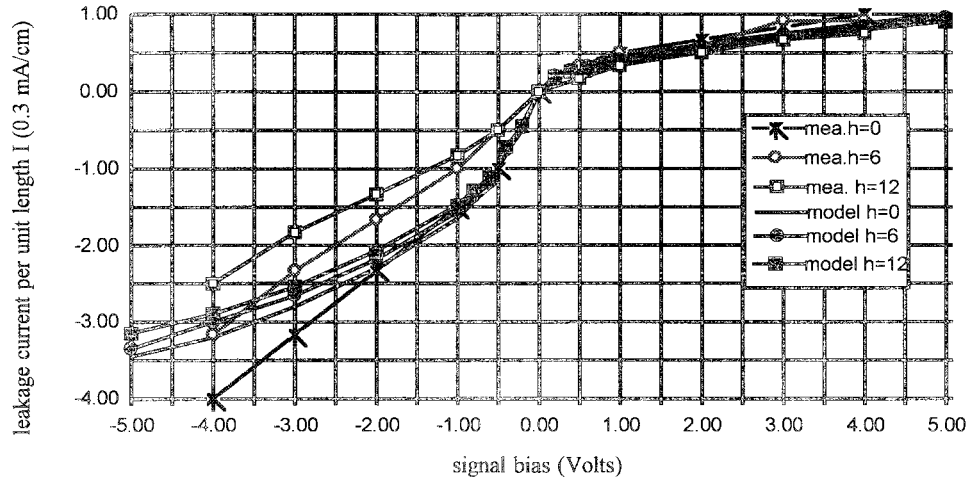
Fig. 6. Simulated and measured I - V curves for different trench CPW lines.

TABLE I
SIMULATED PARAMETERS. HERE, GSG VOLTAGE = 2.0 V,
 $Z_0 = 53.8 \Omega$ BASED IN QUASI-STATIC ANALYSIS [8], [11]

Trench depth h (μm)	0	3	6	9	12
C_t (PF/cm)	1.42	1.26	1.18	1.10	1.04
I_t ($\mu\text{A}/\text{cm}$)	167	162	157	150	137
G_d ($10^{-5} \Omega^{-1}/\text{cm}$)	6.67	6.46	6.33	6.02	5.92
Z_c/Z_0	1	1.06	1.10	1.14	1.17

mapping techniques [9], [13]. The total leakage conduction current is obtained from the surface integral of the modeled current flow distribution. The dynamic shunt conductance is evaluated from the derivative of signal voltage with respect to the induced variation of leakage current. Finally, the normalized characteristic impedance for different trench CPW lines is estimated by the formula $Z_h/Z_0 = \sqrt{C_0/C_h}$, where Z_0 , C_0 refer to the characteristic impedance and capacitance for the nontrenched CPW line with the effective dielectric constant $\epsilon_r = 1$. As this equation comes from a quasi-static consideration [9] and uses the static capacitance, it is a lossless line estimation and, therefore, only an approximation. As can be seen from Table I, the distributed capacitance, leakage conduction current, and shunt conductance have been shown to be reduced, owing to the trench effects. The characteristic

impedance increases by a factor of 1.17 as the trench depth increases from 0 to 12 μm .

III. MEASUREMENT

I - V measurements of CPW lines fabricated according to the previous specifications were carried out by use of an HP 41424A modular dc source/monitor and cascade wafer probe station (typical curves being shown in Fig. 6). The leakage currents are reduced as the trench depth increased, and the measured reverse saturation currents are comparable with the simulated results. These nonlinear characteristics can be explained by the existence of two back-to-back diodes below the signal and ground plate, as shown in Fig. 1. Here, the leakage current path is completed through Schottky diodes $D_{g1,2}$, D_s , and bulk silicon resistor R . It should be remembered that although the area of the $D_{g1,2}$ Schottky diode is much greater than the D_s center diode, the nonlinear leakage current in the electric path is mainly influenced by the reverse biased diode.

CV curves for the CPW line were obtained by use of a wafer probe station and an HP 4275A multifrequency inductance, capacitance, and resistance (ICR) meter at an RF (10-MHz) amplitude of 0.01 V (peak-to-peak). The results (shown in Fig. 7) show the same basic trend as the simulations in Fig. 4.

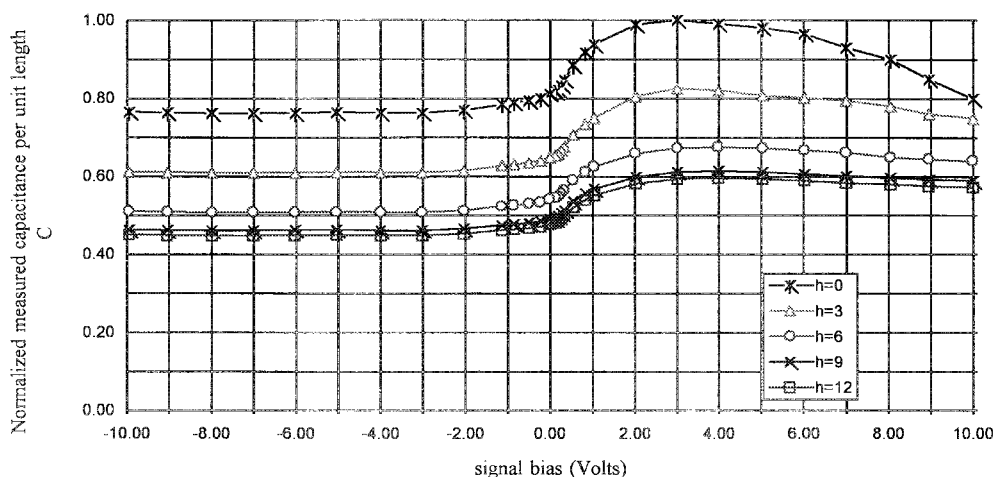


Fig. 7. Measured CV curves in different trenched CPW lines.

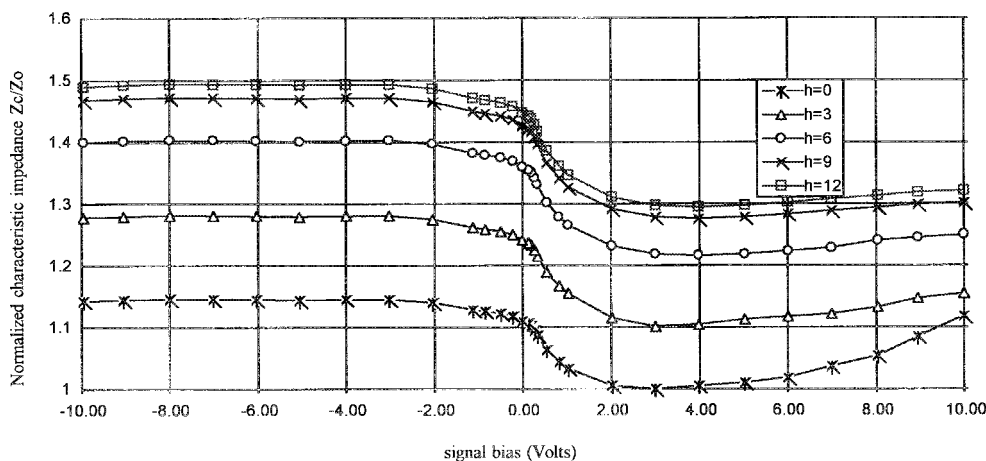


Fig. 8. Characteristic impedance extracted from measured CV data for different trenched CPW lines.

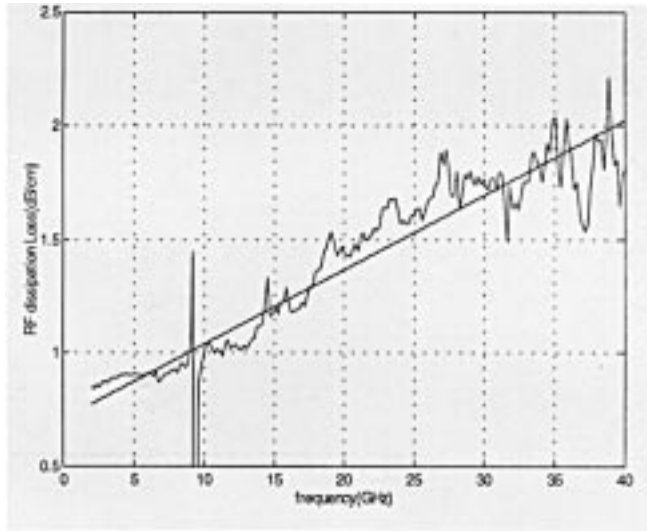
TABLE II
MEASURED RF PARAMETERS VERSUS SIGNAL BIAS FOR A CPW LINE ($h = 0$) BASED ON HRS
($\rho = 10 \text{ k}\Omega\text{cm}$) SILICON 4-IN WAFER AT SPOT FREQUENCY OF 30 GHz

GSG (Volts)	0.0	1.0	2.0	3.0	4.0
$ S_{11} $	0.1036	0.1036	0.1033	0.1036	0.1033
$ S_{21} $	0.8682	0.8783	0.8811	0.8818	0.8826
RF loss (dB/cm)	1.94	1.77	1.73	1.71	1.70
$ Z_c $ (Ohms)	49.0	49.0	49.1	49.2	49.2

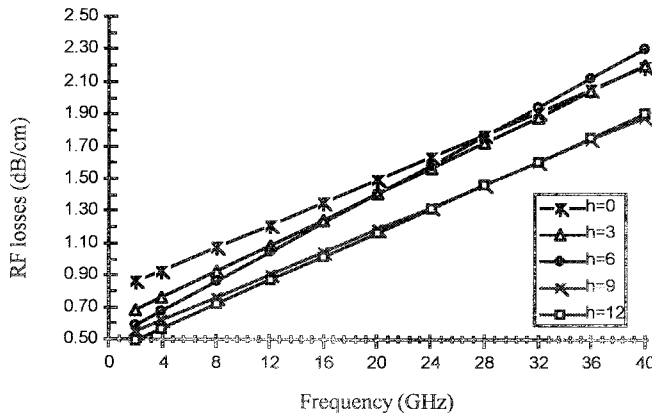
However, the peaks on curves in Fig. 7 are shifted to the forward bias direction compared with those in Fig. 4. This phenomena is under further examination. Fig. 8 shows the corresponding characteristic impedance evaluated from the measured capacitances. It can be seen from Figs. 7 and 8 that the equivalent capacitance presented to the propagating wave tends to decrease after the lines are trenched, and the characteristic impedance for 12- μm trenched CPW lines increases roughly by a factor of 1.3 compared to that of nontrenched lines.

Two-port scattering parameters were measured using an HP 8510B network analyzer and cascade wafer probe station with 200- μm GSG coplanar probes calibrated using line-reflect-match (LRM) to the probe tips, taking care to eliminate multireflection losses [14].² From the measured scattering parameters of the RF dissipation losses, characteristic impedances of the lines have been evaluated. Typical calculated data are shown in Fig. 9(a)–(c). As can be seen in Fig. 9, without trenches, RF losses are 1.2 dB/cm at 10

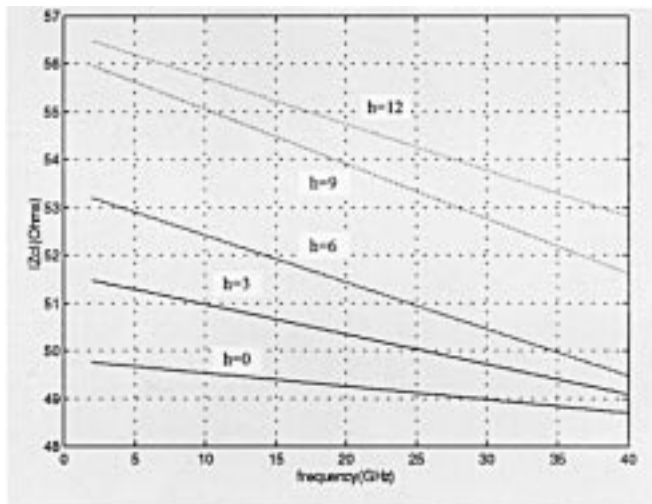
²The RF Capacitor Handbook, American Technical Ceramics, 1979.



(a)



(b)



(c)

Fig. 9. (a) Typical calculated and linear fitted RF dissipation losses for a trench CPW line ($h = 3 \mu\text{m}$). (b) Linear fitted RF losses for the trench CPW lines. (c) Calculated characteristic impedances from measured S -parameters for the trench CPW lines.

GHz and 1.9 dB/cm at 30 GHz. With 9- μm -deep trenches, the line dissipation losses are reduced to 0.8 dB/cm at 10 GHz and 1.6 dB/cm at 30 GHz. Finally, Table II shows the typical RF

losses at different signal bias for a spot frequency of 30 GHz. As can be seen, the losses for zero-bias condition is about 0.2–0.3 dB/cm higher than that for the 4.0-V bias level.

The reduction of RF losses as the signal bias increases may be quantitatively explained by consideration of Fig. 5, in which all of the dynamic shunt conductances in the different trench CPW lines reduce as the signal bias is increased in the positive direction.

IV. CONCLUSIONS

For the first time, the characteristics of trench CPW's for Si MMIC application have been established both experimentally and by semiconductor device simulation. By introducing a vertical trench in the gaps between signal and ground plate, and by dc biasing the CPW line, RF losses can be reduced. This reduction is primarily due to the smaller substrate leakage conduction current and to the reduction in conductor loss owing to the removal of field concentration from the vicinity of the conductor edges. The proposed waveguide structure may be utilized in a special process designed for RF/microwave applications.

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